Qatar-10b

R-0135

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_Qatar-10_230724` · created 2026-07-16T06:39:04+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2460149.9189 +/- 0.0057 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.174 +/- 0.016
Transit depth (Rp/Rs)^2	3.02 +/- 0.57 [%]
Orbital Inclination (inc)	80.94 +/- 2.55
Airmass coefficient 1 (a1)	0.843 +/- 0.058
Airmass coefficient 2 (a2)	0.088 +/- 0.036
Scatter in the residuals of the lightcurve fit is	6.77 %
Best Comparison Star	#3 - [426, 209]
Optimal Aperture	2.05
Optimal Annulus	8.88
Transit Duration (day)	0.082 +/- 0.023

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.174 ± 0.016 vs 0.1265 (deviation 1.28σ)
Duration fit vs published	0.082 d vs 0.1154 d (deviation 0.63σ)
Mid-time O–C	20.3 min
Depth SNR	4.7
Residual scatter	6.77 %

Light curve

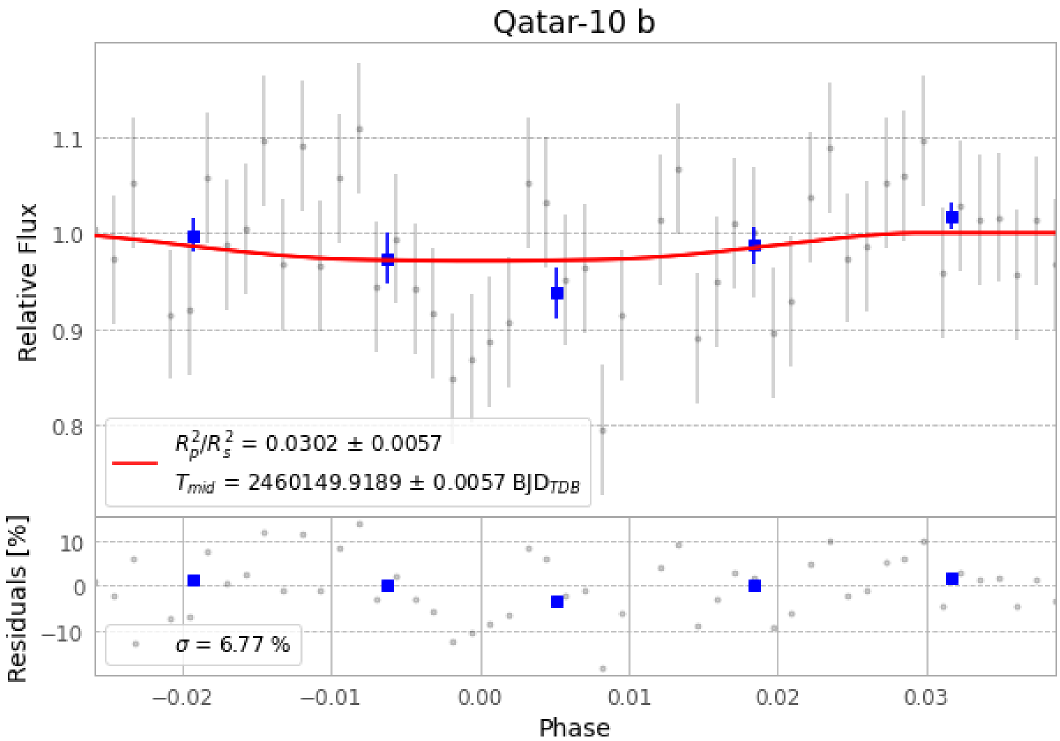

Figure 1 — FinalLightCurve_Qatar-10 b_2023-07-24.png (SHA-256 ee4671cfbb268479…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [290, 191], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 acdb5704bbd0a15f…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 3ebc8ba`

Data provenance

53 FITS frames (35 MB), `Qatar-10230724085417.FITS` → `Qatar-10230724113316.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-10-230724.log (a52ddd75a993…), FinalLightCurve_Qatar-10 b_2023-07-24.png (ee4671cfbb26…), FinalLightCurve_Qatar-10 b_2023-07-24.csv (2c6bf7c6ca91…)

Compute resources

Wall time 175.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)